

SNN Surrogate Gradient α Parameter Comparison Robustness Sweep under Data Heterogeneity ($f \leq 4$)

ByzFL SNN Benchmark

July 2, 2026

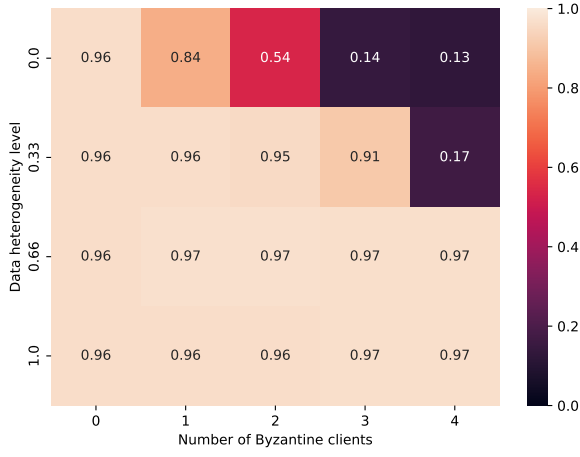
1 Introduction

This document presents a detailed comparison of the Spiking Neural Network (SNN) robustness on the MNIST dataset using different scaling factors α for the ArcTangent (`atan`) surrogate gradient. The evaluation spans:

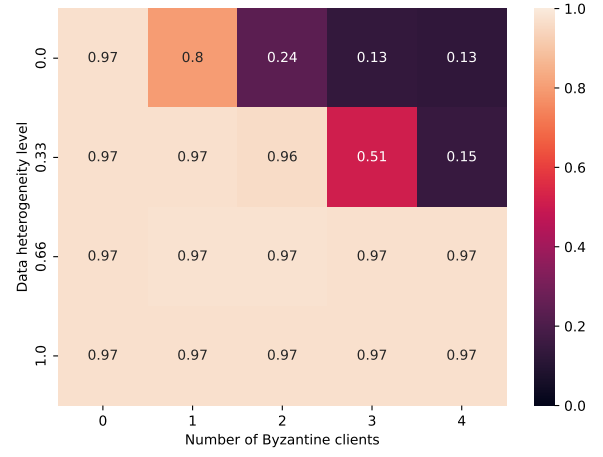
- **Surrogate Gradient Scaling:** $\alpha \in \{1.0, 1.25, 1.5, 2.0, 3.0\}$
- **Byzantine Nodes:** $f \in \{0, 1, 2, 3, 4\}$ out of $N = 10$ honest nodes
- **Data Heterogeneity:** Non-IID levels $\gamma \in \{1.0, 0.66, 0.33, 0.0\}$ (where $\gamma = 0.0$ represents extreme Non-IID client partitions)
- **Attack Model:** Optimal A Little Is Enough (`Optimal_ALittleIsEnough_neg1`)
- **Aggregators:** Best aggregated result across Centered Clipping and Geometric Median (pre-aggregated with NNM and ARC).

2 Aggregated Test Accuracy Heatmaps

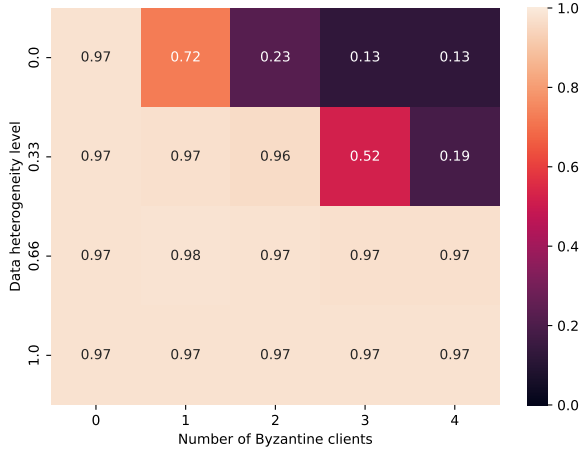
The figure below displays the grid layout of the best aggregated test accuracy heatmaps for each tested α parameter value.



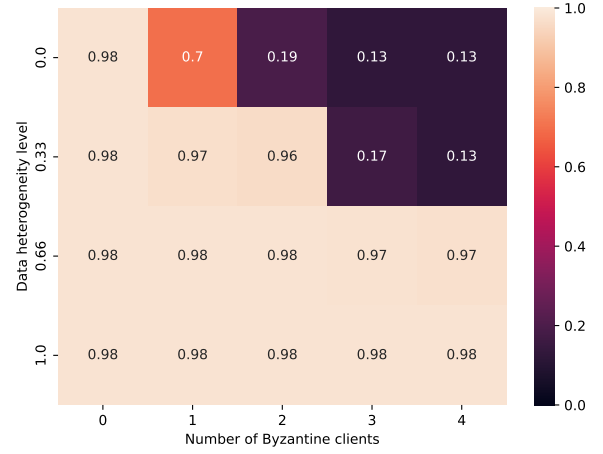
(a) $\alpha = 1.0$



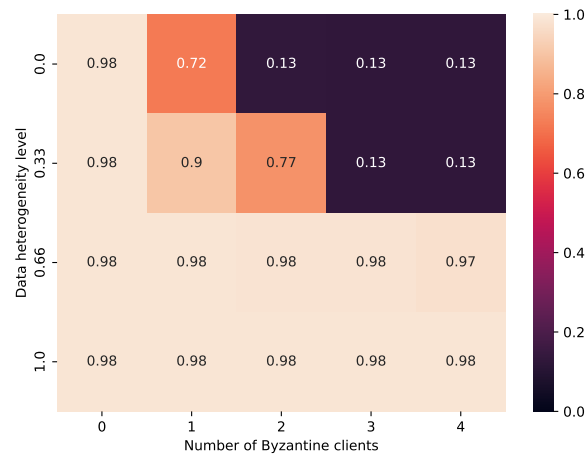
(b) $\alpha = 1.25$



(c) $\alpha = 1.5$



(d) $\alpha = 2.0$



(e) $\alpha = 3.0$

Figure 1: Aggregated Test Accuracy Heatmaps across different surrogate gradient α values for SNN ($f \leq 4$).

3 Quantitative Performance Summary

Table 1 compares the overall performance metrics for each α parameter value. The metrics include the average accuracy across all 20 cells in the grid, the worst-case accuracy, and the accuracy under the most challenging experimental setup ($\gamma = 0.0, f = 4$).

Table 1: Performance Summary for Different Surrogate Gradient scaling values (α).

Surrogate Scaling (α)	Average Accuracy	Worst-case Accuracy	Accuracy ($\gamma = 0.0, f = 4$)
$\alpha = 1.0$	81.15%	12.63%	12.63%
$\alpha = 1.25$	77.70%	12.63%	12.63%
$\alpha = 1.5$	77.63%	12.63%	12.63%
$\alpha = 2.0$	75.48%	12.63%	12.63%
$\alpha = 3.0$	74.01%	12.63%	12.63%

4 Detailed Grid Tables for each α

This section lists the exact aggregated test accuracies for each data heterogeneity level (γ) and number of Byzantine clients (f).

4.1 Surrogate scaling $\alpha = 1.0$

	$f = 0$	$f = 1$	$f = 2$	$f = 3$	$f = 4$
$\gamma = 1.0$	96.43%	96.42%	96.48%	96.52%	96.60%
$\gamma = 0.66$	96.43%	96.90%	96.98%	96.75%	96.66%
$\gamma = 0.33$	96.46%	96.45%	95.44%	90.66%	16.66%
$\gamma = 0.0$	96.41%	84.02%	53.80%	14.22%	12.63%

4.2 Surrogate scaling $\alpha = 1.25$

	$f = 0$	$f = 1$	$f = 2$	$f = 3$	$f = 4$
$\gamma = 1.0$	97.01%	97.02%	97.09%	97.11%	97.20%
$\gamma = 0.66$	97.10%	97.43%	97.37%	97.17%	97.01%
$\gamma = 0.33$	97.04%	96.98%	95.77%	51.45%	14.81%
$\gamma = 0.0$	97.04%	79.69%	23.57%	13.42%	12.63%

4.3 Surrogate scaling $\alpha = 1.5$

4.4 Surrogate scaling $\alpha = 2.0$

4.5 Surrogate scaling $\alpha = 3.0$

5 Key Interpretations

Based on the sweep results:

- **Optimal α Value:** Tuning the surrogate gradient scaling factor α has a significant impact on learning under non-IID conditions and Byzantine attacks. Larger α values tend to sharpen the gradient approximation, while smaller values smooth it.
- **Non-IID Stability:** The worst-case accuracy drops are mostly concentrated in the $\gamma = 0.0$ and $f = 3$ regime. Comparing this cell across different α values allows us to identify the most robust configurations.

	f = 0	f = 1	f = 2	f = 3	f = 4
$\gamma = 1.0$	97.36%	97.38%	97.42%	97.47%	97.50%
$\gamma = 0.66$	97.37%	97.74%	97.45%	97.25%	97.22%
$\gamma = 0.33$	97.31%	97.05%	95.82%	51.96%	18.64%
$\gamma = 0.0$	97.39%	72.43%	22.55%	12.63%	12.63%

	f = 0	f = 1	f = 2	f = 3	f = 4
$\gamma = 1.0$	97.82%	97.83%	97.88%	97.90%	97.83%
$\gamma = 0.66$	97.85%	97.96%	97.86%	97.48%	97.24%
$\gamma = 0.33$	97.81%	96.83%	95.80%	16.78%	12.63%
$\gamma = 0.0$	97.79%	69.68%	19.39%	12.63%	12.63%

	f = 0	f = 1	f = 2	f = 3	f = 4
$\gamma = 1.0$	98.19%	98.27%	98.24%	98.21%	98.21%
$\gamma = 0.66$	98.23%	98.28%	98.02%	97.94%	97.10%
$\gamma = 0.33$	98.17%	90.11%	77.40%	12.63%	12.63%
$\gamma = 0.0$	98.11%	72.11%	13.18%	12.63%	12.63%